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REVIEW ARTICLE

Gut microbiome and human health under the space environment

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Abstract

The gut microbiome is well recognized to have a pivotal role in regulation of the health and behaviour of the host, affecting digestion, metabolism, immunity, and has been linked to changes in bones, muscles and the brain, to name a few. However, the impact of microgravity environment on gut bacteria is not well understood. In space environments, astronauts face several health issues including stress, high iron diet, radiation and being in a closed system during extended space missions. Herein, we discuss the role of gut bacteria in the space environment, in relation to factors such as microgravity, radiation and diet. Gut bacteria may exact their effects by synthesis of molecules, their absorption, and through physiological effects on the host. Moreover we deliberate the role of these challenges in the dysbiosis of the human microbiota and possible dysregulation of the immune system.

Introduction

The gut microbiome is well recognized to have a pivotal role in the regulation of the health and behaviour of its' host (Heyde and Ruder 2015). Recently, there has been a plethora of studies on the human microbiome. Indeed, the human body harbours 10 times greater number of bacteria in comparison to all other types of body cells; and their collective genome (i.e. the microbiome) is much bigger than the human genome (Quigley 2013). Studies have shown that the microbiome is known to guard against a range of disorders such as inflammatory bowel disease, systemic metabolic disease (type 2 diabetes and obesity), atopic eczema and allergic disease (Hoyert *et al.* 2006; Louis, Hold, and Flint 2014; Kang *et al.* 2017; Kelly *et al.* 2017). Moreover research has revealed that the microbiota provides protection against neurodevelopment disorders like autism spectrum disorders (ASD) and schizophrenia (Kang *et al.* 2017; Kelly *et al.* 2017), while others have reported the beneficial properties against cancer (Louis *et al.* 2014; Zhou *et al.* 2017; Roberts *et al.* 2018).

Since 1961, hundreds of astronauts have ventured into space, providing various insights into the effects of several environmental factors such as exposure to radiation, microgravity and the physiological responses of the human body and their consequences on function (Barrett *et al.* 2008). However, the gut microbiome and its role in maintaining astronauts' health has garnered insufficient consideration (Saei and Barzegari 2012). Prior research has been attentive to other aspects of astronaut health such as the musculoskeletal and cardiovascular system, but the precise impact of spaceflight on the gut microbiome and its' effects on human health remains to be determined. Moreover the effects of microgravity on gut microbiome of astronauts are relatively unknown.

The term "microgravity" denotes the setting of "weightlessness" or "zero-g" that are merely endured in the space environment. Moreover, it is noteworthy that the gravitational forces are not wholly equal to zero but are very minute, thus the term "microgravity" is specified. Normally, the "microgravity" level fluctuates from approximately 10^{-3} to 10^{-6} g and depends on the location within the spacecraft and other factors (Huang *et al.* 2018). Thus, the atmosphere of space can affect mammalian health, comprising functions known to interact with the gut microbiota in a bidirectional manner.

Table 1 Effects of microgravity environment on gut microbiota, immune system and other activities

S. no	Pre-flight	During flight	Reference
1.	Sampling began 240 days before flight to establish a baseline of microbiome variability and content	Microbial communities of the gastrointestinal tract, skin, nose and tongue changed during the space mission The composition of the intestinal microbiota became more similar across astronauts in space, mostly due to a drop in the abundance of a few bacterial taxa	Voorhies *et al*. (2019)
2.	Identical twin astronauts went for NASA's first 1-year mission to compare the impact of the spaceflight environment on one twin to the s influence of the Earth environment	Gut microbiota composition was changed significantly Significant changes in gene regulation and telomere length were observed between the twin on earth and astronaut twin Other changes observed include; body weight, sub-foveal choroidal thickness, carotid artery dimensions, serum metabolites and peripapillary total retinal thickness	Garrett-Bakelman *et al*. (2019)
3.	Gut microbiota of astronauts and humans on the earth was studied	The composition of gut microbiota of astronaut's is significantly different (significantly decreased) compared to humans on the earth because of their different lifestyle and dietary structure	Hao *et al*. (2018)
4.	The human microbiome was normal	Disrupt the composition of the human microbiome causing dysbiosis	Voorhies and Lorenzi (2016)
5.	The Murine colon microbiota was normal	Low LET radiation, IRON and spaceflight increased *Bacteroidetes* and decreased *Firmicutes* Spaceflight increased *Clostridiales* and decreased *Lactobacillales*	Ritchie *et al*. (2015)
6.	Strength and muscle volumes were measured from International Space Station (ISS) crewmembers	Muscle volume and strength were decreased in the lower extremities of crewmembers during long-duration spaceflight on ISS	Gopalakrishnan *et al*. (2010)
7.	The pathogenic *Salmonella* in mice gut were studied in spaceflight	Spaceflight-induced increases in *Salmonella* virulence due to simulated microgravity	Wilson *et al*. (2008)
8.	The study investigated debris collected from the crew quarters of the ISS The cultured bacterial counts were more in the ISS samples than cultured fungal population	Crew members were exposed constantly to the dust in a closed system, and also reported allergies The results also exhibited presence of representative of human associated (skin, gastrointestinal track, wounds, etc.) bacterial commensals and opportunistic fungal pathogens	Venkateswaran *et al*. (2014)
9.	The normal microbiota was exposed to radiation effects	Luminal microbiota composition may change that ultimately influence the host's intestinal response to radiation	Packey and Ciorba (2010)
10.	Combined effects of high dietary iron and radiation exposure was assessed	Combined treatments increased liver glutathione peroxidase, serum catalase, and colon myeloperoxidase whereas reducing the concentration of total fecal short-chain fatty acid High-iron diet alone increased leukocyte count Radiation decreased the T-cell CD4:CD8 ratio	Morgan *et al*. (2014)
11.	Mice microbiota at the ground was compared with space travelled mice	Mice microbiota was drastically decreased/altered during space travel	Jiang *et al*. (2019)
12.	Skin fungal microbiota of human was compressively studied in astronauts during a half-year stay at the International Space Station and human on the earth environment	Other fungal diversity was condensed, but the ratio of *Malassezia* to all fungal colonization increased during the astronauts' stay at the ISS	Sugita *et al*. (2016)
13.	Effect of microgravity was assessed on biofilm formation, virulence factors and resistance pattern of *Pseudomonas aeruginosa*, *Salmonella* Typhimurium and *Candida* species	Microgravity affects the biofilm formation, virulence factors and resistance to antimicrobials among *Salmonella* Typhimurium, *P. aeruginosa* and *Candida* species	Wilson *et al*. (2007), Crabbé *et al*. (2010), Altenburg *et al*. (2008)

(Continued)

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Table 1 (Continued)

S. no	Pre-flight	During flight	Reference
14.	Microgravity and space flight effects were studied on flies in space environment	*Drosophila melanogaster* have shown immune disorders (immunocompromised) upon spaceflight and microgravity	Gilbert *et a.l* (2020)
15.	Microgravity and space stress affect the immune system	Immunological responses and immune system in astronauts are harshly suppressed	Aviles *et al*. (2005)
16.	Space environments affect the immune system	Space exposure caused the dysfunction of the immune system	Wilson *et al*. (2007)
17.	Spaceflight encourage microbial growth compared to ground controls	*Escherichia coli* exhibited higher growth (88%) during flight compared to ground controls	Nickerson *et al*. (2004)
18.	Stress effects on human microbiome	Exposure to a space or social stressor changes the number and the composition of gut microbiome	Bailey *et al*. (2011)
19.	Simulated microgravity affects the immune system	Microgravity dysregulate the immune system and the latent human borne viruses reactivated causing human health threat	Stowe *et al*. (2001), Mehta *et al*. (2014)

Various environmental factors that occur in space, including radiation, microgravity and diet are believed to vary the gut microbiota, indicating a risk to the astronaut's health, particularly during long-term spaceflight missions, as listed in Table 1 (Voorhies and Lorenzi 2016; Jiang *et al*. 2019). Astronauts during space travel are highly exposed to different environmental inputs than that of individuals on earth (Ritchie *et al*. 2015).

Microgravity affects gut microbiome composition

Recent studies were conducted by the national aeronautics and space administration (NASA) where the gut microbiome of an astronaut and his twin who remained on Earth were compared. Modifications in the gut microbiome during the 1-year mission onboard the International Space Station (ISS) were detected, that were not seen in his twin brother on Earth during the same time period (Garrett-Bakelman *et al*. 2019). Several changes in microbial community function and composition were apparent during the flight period for the astronaut. The health risks associated with these changes to the gut microbiome are currently not known.

This study was followed up with another study where faecal samples from mice onboard the ISS along with three control groups on Earth during NASA's mission were analysed. Several changes were observed and identified in the gut microbiome due to spaceflight. These spaceflight-associated changes were connected to an altered transcriptome in the liver of the same animals (Jiang *et al*. 2019). In another study, space travellers were sampled pre- spaceflight, during and post-space flight and the results indicated that in the gastrointestinal tract, the overall number of microbes increased while their diversity went down. Alongside, the microbial count of *Serratia marcescens* and *Staphylococcus aureus* (pathogenic bacteria) was highly increased after space travel. Moreover *S. aureus* was transmitted among the astronauts indicating that pathogens can be transferred from individual to individual in a spaceship environment (Voorhies and Lorenzi 2016). Furthermore, several studies have revealed that even short space travel may alter the composition of human microbiota (Saei and Barzegari 2012). These studies are indicative that there are modifications in the gut microbiome of astronauts, following space travel, nonetheless the effects of these changes on the gut microbiome and on the overall health of space travellers are not yet well understood. Furthermore, there is a need to look at the metabolites produced by the gut microbiota of astronauts' following space travel. Once we understand changes in the metabolites of microbiome as a result of microgravity environment, we will be able to propose measures to include prebiotics, probiotics and postbiotics to counter pathological effects in astronauts.

Spaceflight and gut microbiome diversity

To maintain a healthy and diverse consortium of microorganisms, fresh and new microbes are continuously replenishing our body microbiota (Consortium 2012; David *et al*. 2014). The greater the number and types of organism taken in, the healthier the microbiota are (Hanski *et al*. 2012). Healthy adult humans characteristically harbour more than 1000 species of bacteria comprising *Bacteroidetes* and *Firmicutes* being the dominant phyla, and the microbiota is diverse (Shreiner *et al*. 2015). During space travel, there are limited chances of microbial

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replenishment in astronauts as compared to individuals on Earth (Voorhies and Lorenzi 2016). Studies carried out during and after the mission to the ISS revealed that there were changes to gut microbiota diversity in astronauts (Voorhies *et al.* 2019). For each of the crew members, the core microbiome was determined and described. Further analysis revealed 17 gastrointestinal genera whose abundance significantly changed in space. It was revealed that 13 of 17 genera were from the Phylum *Firmicutes*, mainly to the order *Clostridiales*, and nine genera were from the core microbiota of the GI. Furthermore, it was observed that there was a greater than five-fold inflight reduction in *Ruminococcus and Akkermansia*, and approximately three-fold drop in *Fusicatenibacter* and *Pseudobutyrivibrio*. Nonetheless, most of these compositional variations returned to pre-flight levels after the astronauts returned to Earth, with the exemption of two genera of the phylum *Firmicutes* (Voorhies *et al.* 2019). Future studies are warranted to determine the precise effects of these changes in microbiome diversity on astronaut health.

Radiation may alter gut homeostasis

Astronauts during space travel are highly exposed to different environmental inputs than that of individuals on earth (Ritchie *et al.* 2015). Moreover astronauts that travel beyond the low Earth orbit are at the risk of exposure to high linear energy transfer ionizing radiation, primarily due to solar emissions, solar particle events and galactic cosmic rays (Casero *et al.* 2017). Previous work has indicated that radiation that has numerous deleterious effects, that is, increased risk of accelerated bone loss, cataracts and heart diseases (Hamilton *et al.* 2006; Frey 2009).

Prolonged exposure to radiation may significantly induce alterations in gut homeostasis. However, alterations in human gut microbiota through radiation and their role in health risks have not been fully characterized (Casero *et al.* 2017). Several studies revealed that radiation during space travel has significant effects on the intestinal microbiota, as well as the composition of these commensal microbial consortia influences radiation induced-lethality. For example, radiotherapy is done for patients with gastrointestinal problems including cancer. Radiation doses significantly affect gastrointestinal microbial consortia. Similarly, astronauts during space travel are exposed to higher doses of radiation that causes dysbiosis in similar way (McLaughlin *et al.* 1964, Packey and Ciorba 2010).

Diet and the gut microbiome

Several studies have shown that diet may have positive effects on the human intestinal microbiota (Bolnick *et al.* 2014; Paul *et al.* 2015). Consumption of a diet with high dietary iron, in combination with linear energy transfer radiation, which is typical for astronauts, can change faecal concentration signifying the alterations to the composition of the gut microbiota (Morgan *et al.* 2014). The typical diet of astronauts consists of thermostabilized food items such as desserts, entrees, pouched soups, and puddings. Other food items include irradiated meat, rehydratable foods, that is, cereals and spicy green beans, low moisture food items like almonds, peanuts and dried fruit. Bread products with prolonged shelf life like dinner rolls, scones, waffles and tortillas could be formulated and packaged (Cooper *et al.* 2011).

It is recognized that there are two groups of bacteria that reside in the human gut, namely the *Bacteroidetes* and the *Firmicutes* (Ley *et al.* 2006). Ley *et al.* depicted that the relative proportion of *Bacteroidetes* is lower in obese people in comparison to in lean people, and that this proportion increases with weight loss on two types of low-calorie diet, thus representing the significance of diet on the gut microbiota. Other studies also reveal that the composition of the intestinal microbiota is strongly influenced by diet. It has been shown that a high-sugar and high-fat "Western-style" diet augments the relative abundance of *Firmicutes* at the expense of *Bacteroidetes* in animal models (Turnbaugh *et al.* 2008). As *Firmicutes* are needed to absorb fats, higher fat diets can cause an increase in their abundance, leading to weight gain. The study on the effects of long-duration space missions at the ISS on the astronaut microbiome, revealed that 17 gastrointestinal genera whose abundance significantly changed in space. Thirteen out of the 17 genera were *Firmicutes* (Voorhies *et al.* 2019).

Space travel affects the skin and immune system

A plethora of micro-organisms, that is, bacteria, viruses, fungi and parasites inhabit human skin, mucosal surfaces and gut to maintain a fine balance and protect the human body (Findley *et al.* 2013; Kamada *et al.* 2013). The skin and intestinal tract microbiota help in promoting human health by preventing various diseases. These microbes prevent other exogenous pathogens to establish physically, through biofilms on skin surfaces, by maintaining a weakly acidic pH and biologically through producing antimicrobials (Cogen *et al.* 2008; Kamada *et al.* 2013; Christensen and Brüggemann 2014). These microorganisms are thought to have been in a mutual relationship for millions of years with their host, where microbiota contributes in many physiological processes such as the fermentation of carbohydrates, vitamins production and prevent the invasion and colonization by the pathogenic microbes (Hooper and Macpherson 2010; Stecher

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and Hardt 2011; Renz *et al.* 2012). The disruption of the skin microbiota and different host factors may lead the development of skin diseases (Byrd *et al.* 2018). Therefore, it is vita; for astronauts to maintain balance of the skin microbiota to prevent infections during long stays in the ISSs (Sugita *et al.* 2016). Specific human skin borne microbes aid significant contribution to the astronaut's microbiota compared to the human living on earth (Hao *et al.* 2018). For example, the *Malassezia* genus of fungus are abundantly found on human body surfaces. This microbe needs the oily secretion of the sebaceous glands present in mammals. *Malassezia* produces lipase that decompose the glycerine and fatty acids which are important growth factors not only for this fungus but also for other micro-organism inhabit the skin (Hao *et al.* 2018).

Microgravity is among the critical factors that disturb the skin and gut microbiota stability, which may change the microbial gene expression in space environments (Nickerson *et al.* 2004; Voorhies and Lorenzi 2016). For example, microgravity may affect the development of biofilm formation, production of virulence factors and resistance to antimicrobials among, *Pseudomonas aeruginosa*, *Salmonella* Typhimurium and *Candida* species (Wilson *et al.* 2007; Altenburg *et al.* 2008; Crabbé *et al.* 2010). It has been shown that astronauts with sebum on their skin surface during their stay at the space stations have higher levels of *Malassezia* (i.e. production of lipases) than the baseline (Sugita *et al.* 2016). Moreover astronauts have shown disorders in their immune system upon spaceflight (Gilbert *et al.* 2020).

The gut microbiota also plays a critical role in the regulation of the intestinal immune system as shown in (Fig. 1). In spaceflight, the human gut microbiota alters and may cause gastrointestinal and other human diseases (Voorhies and Lorenzi 2016). Similarly, in the human gut, commensal microbes produce several molecules that help the body immune system in the protection against pathogenic micro-organisms, their toxins and other pathobionts (Kamada *et al.* 2013) as shown in (Fig. 2). Intestinal microbiota synthesizes several short-chain fatty acids and bacteriocins that directly prevent pathogen growth. These commensals bacteria suppress the growth of pathogens by their metabolites and in some cases can also change the expression of the virulence factor in pathogens by consuming residual oxygen. Gut microbiota upregulate the mucus layer and thus facilitate host barrier function (Kamada *et al.* 2013). In the absence or disrupted commensal microbiota, the pathogens (*Salmonella* and *Shigella*) and pathobionts (vancomycin-resistant *Enterococci* and *Clostridium difficile*) may get enter into the gut epithelium followed by blood vessels and causes systemic infections (Kamada *et al.* 2013).

Additionally, commensal microbiota activates antigen presenting cells as well as other immune cells which in turn switch on a cascade of immune responses. As a result of this, various types of interleukin (ILs) and cytokines are released that are primarily important to assist human microbiota in pathogen clearance and other inflammatory events (Kamada *et al.* 2013; Belkaid and Hand 2014). In contrast, the immune system during space travel is significantly affected due to space environmental conditions. Studies have revealed that the immune system and immunological responses in astronauts are drastically suppressed (Aviles *et al.* 2005). Stay for long period in space resulted in dysfunction of the immune system and could increase the risk of infectious diseases (Wilson *et al.* 2007). Commensal microbiota successfully outcompetes the pathogenic microbes and inhibit their entry and colonization by utilizing all the nutrients and other growth factors within the human gut. For example, indigenous *Escherichia coli* outcompetes the enterohaemorrhagic *E. coli* for essential minerals, amino acids, organic acids, and other nutrients in the intestine (Momose *et al.* 2008a, 2008b; Leatham *et al.* 2009).

During space travel the astronaut's immune system is not properly functioning (Crucian *et al.* 2013; Cervantes and Hong 2016). For example, the immune system keeps several humans borne latent viruses such as Epstein-Bar virus, herpes simplex virus1 and varicella zoster virus in deactivated form. But the impaired immune system due to microgravity and other space travel stressors these dormant viruses reactivated that may make favourable conditions for their growth thus becoming a probable health risk (Stowe *et al.* 2001; Mehta *et al.* 2014).

Bacterial exposure to spaceflight showed higher resistance toward antibiotics, but this phenomenon is not so obvious, as few studies have been shown that when skin microbiota were exposed to microgravity for long periods on a space station, bacteria exposed become more susceptible to same antibacterials upon return to earth (Juergensmeyer *et al.* 1999). Tixador *et al.* (1985) and Moatti *et al.* (1986) revealed that microbes cultured from astronauts during flight showed more resistance than the bacteria isolated from the same individuals before and after the space flight. They performed experiments to elucidate the antibiotic susceptibility of *E. coli* and *S. aureus* isolated from the GI and nasal microbiota astronaut (Taylor 2015). For *E. coli* the antibiotic dose was almost four times higher while for *S. aureus*, twofold higher dose was recorded at space compared to ground control (Taylor 2015). In another study it was observed that as crew members are exposed constantly to dust in a closed system, allergies were reported. The results from this study also revealed the presence of human associated (skin, gastrointestinal track, wounds, etc.) bacterial commensals

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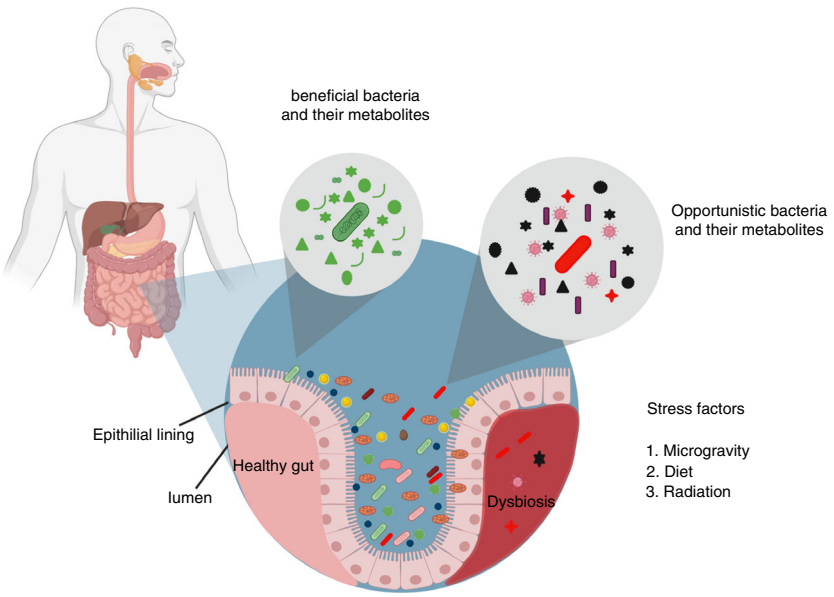

Figure 1 Spaceflight challenges and the gut microbiota. Pre-spaceflight in a healthy human intestine, indigenous microbes assist the immune system and regulate gut homeostasis. In case of disrupted gut microbiota due to microgravity, radiation and other spaceflight stress factors, gut microbiota may breach the gut lining due to environmental stress (space environment) and gain entry to cause systemic infections. (●▲●) beneficial metabolites; (●▲●) harmful metabolites; (▬) pro-inflammatory bacteria; (▬) beneficial bacteria. [Colour figure can be viewed at wileyonlinelibrary.com]

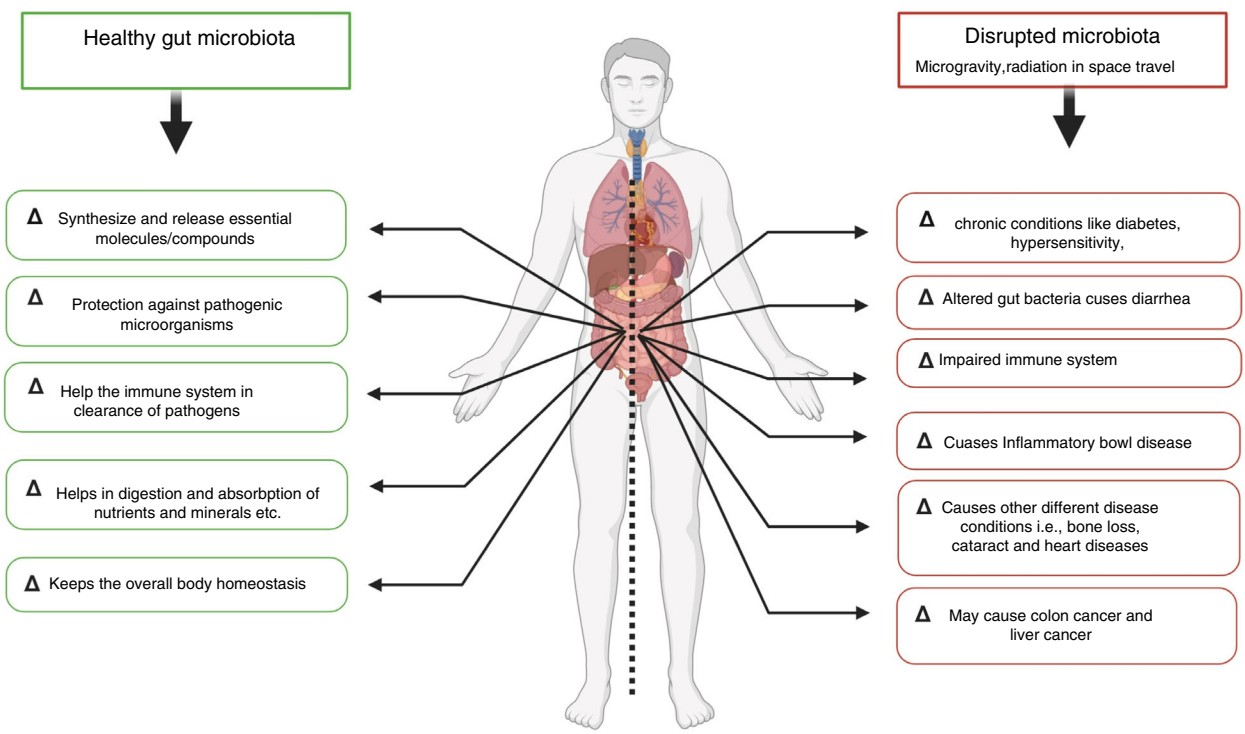

Figure 2 Effects of microgravity, radiation, diet and other environmental factors on gut microbiota during space travel. In case of healthy gut, the microbiota play important role in clearance of pathogenic microbes, maintain body homeostasis, helps the immune system and produces essential metabolites. Microbial imbalance due to space travel could result in several detrimental effects such as impaired immune system, diarrhoea and other chronic diseases, colon cancer. [Colour figure can be viewed at wileyonlinelibrary.com]

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and opportunistic fungal pathogens in found in space station debris and surfaces (Venkateswaran *et al.* 2014).

Space travel may affect commensal bacteria

Bacterial genes and proteins are expressed differently during space travel and on the ground. For example, several genes (of known and unknown functions) and various types of proteins in *Salmonella* Typhimurium were differentially regulated/expressed during spaceflight as compared to the ground cultures of the same bacteria (Wilson *et al.* 2007). Microarray pattern revealed the widespread alterations of gene expression distributed globally throughout the chromosome (Wilson *et al.* 2007). Studies have shown that spaceflight promotes the growth of micro-organisms compared to ground controls (Nickerson *et al.* 2004). For example, *E. coli* exhibited higher growth during flight where the lag phase was shortened followed by the extended log phase, as a result the microbial cell density increased by 88% compared to ground controls (Nickerson *et al.* 2004).

Omics technological advances have allowed the scientific community to carry out genome-wide studies to understand the effects of space travel on microbes. These studies revealed that microgravity altered several microbial characteristics, that is, alter their gene expression, virulence and behaviour. For example, microgravity resulted in the formation of biofilms by opportunistic *S. aureus* which in turn reduces the growth rate and degree of virulence of these bacteria (Castro *et al.* 2011). Similarly, *P. aeruginosa* PAO1, drastically altered the expression of certain gene as well as proteins in modelled microgravity, but changes in their virulence is obscure (Crabbé *et al.* 2011). The pathogenic fungus, *Candida albicans* when grown in space environments changes its budding configuration, cell aggregation and gene expression with no improvement of virulence (Crabbé *et al.* 2013). Furthermore, when *Salmonella enterica serovar* Typhimurium was grown under virtual microgravity, significant increase in cell number in infected mice, its virulence, and its endurance within a macrophage were observed (Nickerson *et al.* 2000; Voorhies and Lorenzi 2016).

Gut microbiota and diseases

Alterations in the gut microbiota are well known to cause several human diseases, including ASD (Mayer *et al.* 2014). Since the gut microbiota contributes in human metabolism, maintenance of immune system, it may also control the activities of the central nervous system through, endocrine, neural and immune pathways (Sampson and Mazmanian 2015), it is assumed that the gut microbiota play a key role in ASD pathophysiology. It is believed that the gut microbiota and the CNS have an indirect link known as "microbiome-gut-brain axis" (Strati *et al.* 2017).

Exposure to space atmosphere induces changes in lipids and glucose metabolism that further causes bone loss, cardiovascular disorders and muscular atrophy in astronaut. The gut microbiota also alters in rodents and humans during pre and post spaceflight for short- and long-term duration (Wang *et al.* 2019). Due to the presence of abundant number of microbes, human has been called a "supra-organism" while the GI flora has been considered as the virtual organ of the body (Saei and Barzegari 2012). Also, gut microbiota regulating the mucosal barrier integrity, host metabolism and the maturation of individual's immune system (Sittipo *et al.* 2018). Fluctuations in gut flora is closely related with several metabolic syndromes such as dyslipidaemia, insulin resistance, obesity, glucose intolerance, and accumulation of fat in the liver (Wang *et al.* 2019).

Microbial imbalance has also been observed during spaceflight. During space flight, beneficial bacteria, that is, *Lactobacillus* and *Bifidobacterium* species reduces in astronauts' body and in contrast, the pathogenic/opportunistic microbes (*Clostridium*, *E. coli, Fusobacterium nucleatum* and *P. aeruginosa*) accumulate and increases their population (Smirnov and Lizko 1987; Ritchie *et al.* 2015; Garrett-Bakelman *et al.* 2019). This shows that space conditions, that is, diet, microgravity, stress and radiation (Mardanov *et al.* 2013), may change the structure as well as composition of gut microbiota. Mice and rats on space flight have shown higher level of liver gluconeogenesis and liver glycogen content (Wang *et al.* 2019).

Stress and human gut microbiome

In the human intestine, the human flora directly effects the first line of resistance (biochemical, immunological and physiological pathways) to numerous diseases (Hecht 1999; Saei and Barzegari 2012). As space travel is a stressful environment (microgravity, radiation and high iron diet), this can trigger alteration in the composition of microbiota and their gene expression which may lead to either transient or permanent governance of pathogenic/opportunistic bacterial species (Dumas *et al.* 2006). Exposure to social stressors changes the number as well as the composition of the gut microbiome, it may also change their stability and could leads to gut bacterial translocation (Bailey *et al.* 2011). Environmental stress also increases the level of circulation of monocyte chemoattractant protein-1 (MCP-1) and interleukin 6 (IL-6) which has a positive link to changes in *Coprococcus, Dorea* and *Pseudobutyrivibrio* bacterial genera in the

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cecum (Bailey *et al.* 2011; Saei and Barzegari 2012). This suggests that the gut microbiota may contribute in stressor-induced immunoenhancement. For example, Bailey *et al.* (2011) treated mice with a cocktail of antibiotics to see whether reducing microbiota would terminate stressor-induced increase in circulating cytokines and the results revealed that mice treated with antibiotics (stress factor) failed to increase IL-6 and MCP-1 circulation confirming that the gut microflora is very important for the observed increase in circulating cytokines (Saei and Barzegari 2012).

Significance and Impact

There is a dearth of studies examining the effects of the space environment on the gut microbiome of space travellers. Recently, three nations including the United Arab Emirates, China and the United States of America are launching rover missions to Mars in July 2020 (Witze *et al.* 2020) with a view for manned missions in the upcoming decades. Long-duration missions that are planned to take humans to Mars and even further are in the pipeline in forthcoming years, thus studying the health of astronauts is of paramount importance and further studies on the gut microbiome are needed to comprehend the reasons for the changes in diversity of the gut microbiome in the space environment. The NASA study on twins revealed several changes in microbial community function and composition, however, the health risks associated with these changes to the gut microbiome are currently not known (Garrett-Bakelman *et al.* 2019). Furthermore, the study had limitations, as only a single astronaut was analysed, thus additional studies with more subjects undergoing long-duration (≥12 months) missions are needed to confirm these findings. The influence that the gut microbiome has on human health is highly significant. A plethora of studies have linked the gut microbiome with neurological and physiological conditions, like Alzheimer's disease, Parkinson's disease, autism and schizophrenia (Kang *et al.* 2017; Kelly *et al.* 2017), while others have reported the beneficial properties against cancer (Louis *et al.* 2014; Zhou *et al.* 2017; Roberts *et al.* 2018). Thus, further studies on the gut microbiome of astronauts are warranted to mitigate risks to crew health and performance during planned future long-term spaceflight missions.

Concluding remarks

Overall, spaceflight has significant effects on the alteration of human microbiota and immune system of astronauts. Space travel changes the physiology as well as the composition of gut microbiome. Several specific environmental factors in space, including microgravity, radiation and diet may lead to fluctuation of the gut microbiota, indicating a health risk to the astronauts, particularly during extended spaceflight missions. Microgravity is a significant factor that effects the diversity and composition of gut microbiota. Radiation has also deleterious effects on astronauts. Moreover extended exposure to radiation may significantly induce alterations in gut homeostasis and microbiota. Omics studies have revealed that real or simulated microgravity alters behaviour, virulence and gene expression in several micro-organisms. Moreover alteration in the gut microbiota resulted in infections due to opportunistic microbes. Further studies are warranted to determine the metabolites produced by the gut bacteria under microgravity environment and compared with metabolites produced in normal environments on earth. Importantly, the effects of these molecules on obesity, ageing, senescence and longevity, *in vitro* and *in vivo* need to be elucidated. It will also be interesting to understand if these molecules have any neuroprotective effects. Elaborating on the understanding of how the gut microbiota and the metabolites produced under microgravity conditions, in comparison to normal environments will be pivotal in the advancement of sufficient measures against the unfavourable changes often seen in astronauts after return. This will be a key step to ensure the efficiency and safety of forthcoming space missions.

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Author contributions

NAK and RS conceived the idea. RS and NA reviewed literature and prepared the first draft of the manuscript. NAK and RS corrected the manuscript. All authors read and approved the final manuscript.

Conflict of Interest

No conflict of interest exists.

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